
Metadata-Free Inference under Packet Loss and Reordering via Implicit Positional Encoding and Content-Addressable Retrieval

Yaswanth Ram Kumar
Department of Electrical Engineering
Indian Institute of Technology Bombay
23b1277iitb.ac.in

Nikhil Karamchandani
Department of Electrical Engineering
Indian Institute of Technology Bombay
nikhilk@ee.iitb.ac.in

Abstract

Split inference deploys the encoder of a neural network on an edge device and the decoder on a remote server, connected by an unreliable communication channel. Intermediate activations are fragmented into packets that may arrive out of order or be lost entirely. Classical solutions attach explicit sequence numbers to each packet, incurring metadata overhead and failing when index information is corrupted. We propose a framework that eliminates all metadata by encoding positional information implicitly into the representation: each activation chunk is masked by a chunk-specific random sign vector, so that the position of a received packet can be inferred by matching against a reference corpus without transmitting any index. We develop a complete theoretical analysis showing that cross-position inner products concentrate tightly around zero while correct matches maintain a positive signal, and derive an explicit recovery condition in terms of chunk dimension, corpus size, and packet loss rate. We formulate position assignment as bipartite matching, solvable in $O(P^3)$ via the Hungarian algorithm, reducing an exponential combinatorial search space of size $P!/(P-k)!$ to polynomial time. We identify two inference modes: retrieval-only prediction directly from corpus labels, and hybrid reconstruction that imputes missing chunks before decoding. Empirically, on MobileNetV2, ResNet50, and ViT-B16, our content-based rematching achieves accuracy identical to oracle alignment at up to 40% packet loss under full reordering, while random alignment collapses to near-chance performance.

1 Introduction

The deployment of deep neural networks on resource-constrained edge devices has motivated a family of systems known as *split inference* [Kang et al., 2017]. A network f is partitioned as $f = f_{\text{dec}} \circ f_{\text{enc}}$, where f_{enc} runs on an edge device and f_{dec} runs on a powerful remote server. The intermediate activation $z = f_{\text{enc}}(x) \in \mathbb{R}^d$ must be transmitted from the edge to the server before the server can evaluate $\hat{y} = f_{\text{dec}}(z)$.

In practice, activations are too large for a single IP packet and must be fragmented into P chunks. Over realistic UDP-based channels, two failure modes arise *simultaneously*:

- **Packet loss:** a subset of chunks does not arrive.
- **Reordering:** packets arrive in an arbitrary order unrelated to their transmission order.

The classical solution attaches explicit sequence numbers to each chunk, allowing the receiver to identify positions and detect gaps. This approach incurs bandwidth overhead, exposes protocol structure, and fails entirely when index metadata is corrupted alongside the payload.

Central question. *Can accurate inference be performed under simultaneous packet loss and reordering, without transmitting any positional metadata, using the semantic content of activations alone?*

We answer affirmatively. Our key observation is that high-dimensional neural activations occupy a structured manifold. By masking each chunk with a position-specific random sign vector, positional identity becomes recoverable from content: the correct chunk position produces a positive inner product with its corpus counterpart, while all incorrect positions produce mean-zero inner products. We then reduce joint corpus identification and position assignment to a two-stage polynomial-time algorithm.

1.1 Summary of Contributions

1. **Problem framing: joint loss and reordering.** We study the harder, more realistic setting where received packets are both incomplete and unordered. The brute-force search space has size $P!/(P-k)!$ (falling factorial), which is $\approx 4.4 \times 10^{11}$ for $P = 16$ at 25% loss. We show this is reducible to polynomial time through representation structure alone, without any protocol overhead.
2. **Implicit positional encoding via random sign masking.** We prove that masking each chunk with a position-specific random sign vector $s_j \in \{-1, +1\}^{d_c}$ makes correct-position inner products preserve the original cosine similarity, while cross-position inner products are exactly mean-zero with variance decaying as $O(1/d_c)$. The sign vectors are shared but never transmitted.
3. **Formal recovery guarantee.** We prove that correct position assignment is achieved with probability at least $1 - P^2 \exp(-c d_c \Delta^2 / (1 + d_c \sigma^2))$, where Δ is the minimum signal margin across chunk positions. This yields an explicit three-way tradeoff between chunk dimension d_c , the number of competing positions P , and channel noise σ^2 .
4. **Two-stage polynomial algorithm via bipartite matching.** We introduce an $O(MP^2)$ aggregation step for corpus identification followed by an $O(P^3)$ Hungarian algorithm for position assignment. Together these replace the exponential $O(P!/(P-k)!)$ brute-force search. A greedy $O(P^2)$ variant suffices when the recovery condition is comfortably satisfied.
5. **Two inference modes.** Retrieval-only mode predicts from corpus labels without running the decoder; hybrid mode imputes missing chunks from the corpus and decodes the reconstructed activation. We characterize when each is appropriate.
6. **Empirical validation.** Rematched accuracy equals oracle on MobileNetV2, ResNet50, ViT-B16 (ImageNet-1K), DistilBERT (SST-2), and BERT-Base (MNLI) under full reordering at up to 40% packet loss.

2 Related Work

Split inference. Neurosurgeon [Kang et al., 2017] studies where to partition a network to minimize end-to-end latency between edge and cloud. BottleNet++ [Shao and Zhang, 2020] compresses intermediate features for bandwidth efficiency in device-edge co-inference. JALAD [Li et al., 2018] jointly optimizes accuracy and latency under communication constraints. These works fix the split and optimize bandwidth or split-point selection; our work fixes the split and addresses transport reliability under loss and reordering.

Robustness to packet loss. SI-NR [Shao et al., 2023] trains the full model end-to-end with dropout-style masking so the decoder tolerates missing activations. This requires separate retraining per target loss regime and does not recover packet positions or content. Our method is zero-shot on any pretrained model and recovers positions entirely from content.

Retrieval-augmented methods. Retrieval-augmented generation [Lewis et al., 2020] augments language model outputs with retrieved documents. Our retrieval operates at the activation level and serves a different purpose: recovering spatial alignment, not augmenting world knowledge.

Contrastive learning. InfoNCE [van den Oord et al., 2018] learns representations where matching pairs have higher similarity than non-matching pairs. We optionally apply it at the chunk level as a fine-tuning stage. The contribution is the problem framing — chunk alignment recovery as a downstream task — not the loss itself.

Content-addressable memory. Hopfield networks [Hopfield, 1982] demonstrate that content can serve as an address in associative memory. Modern continuous Hopfield networks [Ramsauer et al., 2020] achieve exponential storage capacity. Our corpus-based retrieval is a high-dimensional instantiation of this principle.

Assignment problems. The linear assignment problem and its solution via the Hungarian algorithm [Kuhn, 1955] underpin Stage 2 of our method. The key recognition is that packet position recovery in a sign-masked framework reduces exactly to a maximum-weight bipartite matching instance.

High-dimensional probability. Concentration of inner products in high dimensions is classical [Vershynin, 2018]. We apply these tools to derive explicit recovery thresholds for sign-masked activation matching.

3 Problem Formulation

3.1 Setup

Let x be an input to a split network $f = f_{\text{dec}} \circ f_{\text{enc}}$. The edge device computes the activation:

$$a = f_{\text{enc}}(x) \in \mathbb{R}^d.$$

The activation is partitioned into P equal-sized chunks of dimension $d_c = d/P$:

$$a = (a_1, a_2, \dots, a_P), \quad a_j \in \mathbb{R}^{d_c}, \quad \|a_j\|_2 = 1.$$

Assumption 1 (Channel model). *The channel independently drops each of the P packets with probability $p_{\text{loss}} \in [0, 1)$. The arrival order of surviving packets is arbitrary and unknown to the receiver. No packet identifiers, sequence numbers, or timestamps are transmitted.*

The receiver observes $P - k$ packets, where $k \sim \text{Binomial}(P, p_{\text{loss}})$:

$$Q = \{q_1, q_2, \dots, q_{P-k}\}.$$

Each q_i is a noisy version of some activation chunk $a_{\sigma(i)}$, where $\sigma : [P - k] \rightarrow [P]$ is an *unknown injective function* — unknown both in which positions arrived and in what order.

Assumption 2 (Noise model). *Received packets satisfy $q_i = a_{\sigma(i)} + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2 I)$ independently.*

A corpus of M reference activations is available at the server:

$$\mathcal{C} = \{a^{(1)}, \dots, a^{(M)}\}, \quad a^{(m)} = (a_1^{(m)}, \dots, a_P^{(m)}).$$

3.2 Alignment as Combinatorial Optimization

Definition 1 (Correct alignment). *Correct alignment is the joint recovery of (i) the corpus index m^* whose entry is closest to a , and (ii) the injective mapping σ that assigns each received packet q_i to its true chunk position $\sigma(i)$.*

Naively, alignment requires:

$$(\hat{m}, \hat{\sigma}) = \arg \max_{m \in [M], \sigma : [P-k] \hookrightarrow [P]} \sum_{i=1}^{P-k} q_i^\top a_{\sigma(i)}^{(m)},$$

where $\hookrightarrow$ denotes injective maps.

Combinatorial complexity. The number of injective maps from $P - k$ received packets to P positions is the falling factorial:

$$P^{\underline{P-k}} = \frac{P!}{(P - (P - k))!} = \frac{P!}{k!}.$$

For $P = 16$ and 25% loss ($k = 4$): $16!/4! \approx 4.4 \times 10^{11}$. This is completely intractable by brute force.

Comparison with the ordered case. If packets are guaranteed to arrive in transmission order (loss-only, no reordering), the receiver only needs to determine *which* $P - k$ positions survived, not their assignment order. The search space collapses to $\binom{P}{P-k}$, which for the same example is only $\binom{16}{12} = 1820$. Our setting is strictly harder. We address the joint loss-and-reordering case throughout.

4 Method

4.1 Implicit Positional Encoding via Random Sign Masking

We assign each chunk position $j \in [P]$ a fixed random sign vector:

$$s_j \in \{-1, +1\}^{d_c}, \quad s_{j,\ell} \stackrel{\text{iid}}{\sim} \text{Uniform}\{-1, +1\}.$$

The sign vectors are generated once from a shared random seed, stored at both the edge device and the server, and *never transmitted*. Before transmission, the edge applies elementwise masking:

$$\tilde{a}_j = s_j \odot a_j \in \mathbb{R}^{d_c}.$$

The corpus is stored in masked form: $\tilde{a}_j^{(m)} = s_j \odot a_j^{(m)}$.

Correct position — signal preserved. When the server compares received chunk $\tilde{a}_j$ against the corpus at the same position j :

$$\langle \tilde{a}_j, \tilde{a}_j^{(m)} \rangle = \sum_{\ell} s_{j,\ell}^2 a_{j,\ell} a_{j,\ell}^{(m)} = \langle a_j, a_j^{(m)} \rangle \rho_j > 0. \quad (1)$$

The mask cancels exactly ($s_{j,\ell}^2 = 1$) and the original cosine similarity is preserved.

Incorrect position — mean-zero noise. When the server compares $\tilde{a}_j$ against corpus position $j' \neq j$:

$$\langle \tilde{a}_j, \tilde{a}_{j'}^{(m)} \rangle = \sum_{\ell} (s_{j,\ell} s_{j',\ell}) a_{j,\ell} a_{j',\ell}^{(m)}. \quad (2)$$

Since s_j and $s_{j'}$ are independent, each product $s_{j,\ell} s_{j',\ell}$ is ± 1 with equal probability independently of activation values. This makes (2) a sum of mean-zero, bounded random variables — analyzed fully in Section 5.

4.2 Global Permutation

Convolutional feature maps have spatially smooth structure, meaning adjacent chunks a_j and a_{j+1} may be correlated. This inflates cross-position inner products for nearby positions. To remove this residual correlation, we apply a fixed global permutation $\Pi \in \mathfrak{S}_d$ to the flattened activation before chunking:

$$a \leftarrow \Pi a.$$

Π is shared between edge and server and never transmitted. After Π , the mapping from spatial feature map location to chunk assignment is randomized, satisfying the approximate independence condition required by Assumption 3 in Section 5.

4.3 Corpus Construction

The server pre-computes a corpus from a held-out representative dataset $\mathcal{D}_{\text{ref}}$:

$$\mathcal{C} = \{f_{\text{enc}}(x^{(m)})\}_{m=1}^M, \quad x^{(m)} \in \mathcal{D}_{\text{ref}}.$$

Each entry is stored in masked, permuted form. The corpus may use the training set and requires no new data collection. The number of entries M controls the resolution of the non-parametric prior over the activation manifold; sensitivity to M is studied in Section 8.

4.4 Two-Stage Alignment Algorithm

Our algorithm proceeds in two stages: corpus identification via aggregation scoring, and position assignment via bipartite matching.

4.4.1 Stage 1: Corpus Identification via Aggregation Scoring

For each received packet q_i and corpus entry m , compute inner products with all masked corpus chunks:

$$v_{i,m,j} = \langle q_i, \tilde{a}_j^{(m)} \rangle, \quad i \in [P-k], m \in [M], j \in [P].$$

Definition 2 (Aggregation score).

$$\text{score}(m) = \sum_{i=1}^{P-k} \max_{j \in [P]} \langle q_i, \tilde{a}_j^{(m)} \rangle.$$

The $\max_j$ greedily assigns the best-matching position to each received packet independently, then sums the evidence across all surviving packets. The nearest corpus entry is:

$$m^* = \arg \max_{m \in [M]} \text{score}(m).$$

Complexity: $O(M \cdot (P-k) \cdot P) = O(MP^2)$ inner products. For $M = 1000, P = 16, d_c = 64$: approximately 10^7 multiply-accumulate operations, taking $\ll 1$ ms on a modern GPU.

4.4.2 Stage 2: Position Assignment via Bipartite Matching

Given m^* , position assignment is formulated as a **maximum-weight bipartite matching problem**. Construct the benefit matrix $B \in \mathbb{R}^{(P-k) \times P}$:

$$B_{ij} = \langle q_i, \tilde{a}_j^{(m^*)} \rangle, \quad i \in [P-k], j \in [P].$$

We seek the injective assignment $\hat{\sigma} : [P-k] \rightarrow [P]$ maximizing the total matching weight:

$$\hat{\sigma} = \arg \max_{\sigma \text{ injective}} \sum_{i=1}^{P-k} B_{i, \sigma(i)}.$$

This is a linear assignment problem on a bipartite graph with $P-k$ packet nodes on the left and P position nodes on the right. It is solved exactly by the **Hungarian algorithm** [Kuhn, 1955] in $O(P^3)$ time. Since $P-k < P$, the matrix B is wide rather than square; the Hungarian algorithm handles this by adding dummy rows with zero weight.

Greedy variant. When the recovery condition of Theorem 4 is comfortably satisfied (large d_c , low p_{loss}), greedy assignment suffices and avoids the $O(P^3)$ cost:

$$\hat{\sigma}(i) = \arg \max_{j \notin \{\hat{\sigma}(1), \dots, \hat{\sigma}(i-1)\}} B_{ij}, \quad i = 1, \dots, P-k.$$

Complexity: $O(P^2)$. We compare greedy and Hungarian empirically in Ablation 4.

Complexity reduction.

Approach	Search space	Complexity
Brute force	$P!/(P-k)!$	Exponential
Stage 1 + Hungarian	—	$O(MP^2 + P^3)$
Stage 1 + Greedy	—	$O(MP^2 + P^2)$

The full procedure is in Algorithm 1.

4.5 Inference Modes

Mode 1: Retrieval-only. Return the precomputed label of the nearest corpus entry:

$$\hat{y} = y^{(m^*)}.$$

This bypasses the decoder entirely. Appropriate when the corpus is large and fine-grained enough that $x^{(m^*)}$ is semantically close to x .

Algorithm 1 Content-Addressable Alignment for Split Inference

Require: Received packets $\{q_i\}_{i=1}^{P-k}$, masked corpus $\{\tilde{a}^{(m)}\}_{m=1}^M$, sign vectors $\{s_j\}_{j=1}^P$

Ensure: Corpus index m^* , injective assignment $\hat{\sigma} : [P-k] \rightarrow [P]$

Stage 1: Corpus identification

- 1: **for** each $m \in [M]$ **do**
 - 2: $\text{score}(m) \leftarrow \sum_{i=1}^{P-k} \max_{j \in [P]} \langle q_i, \tilde{a}_j^{(m)} \rangle$
 - 3: **end for**
 - 4: $m^* \leftarrow \arg \max_m \text{score}(m)$
-

Stage 2: Position assignment

- 5: Build $B \in \mathbb{R}^{(P-k) \times P}$: $B_{ij} \leftarrow \langle q_i, \tilde{a}_j^{(m^*)} \rangle$
 - 6: $\hat{\sigma} \leftarrow \text{HUNGARIAN}(B)$ // or Greedy when SNR is high
-

Output: $(m^*, \hat{\sigma})$

Mode 2: Hybrid reconstruction and decoding. Unmask received chunks and impute missing positions from the corpus:

$$\hat{a}_j = \begin{cases} s_j \odot q_i & \text{if } j \in \text{range}(\hat{\sigma}) \text{ with } \hat{\sigma}(i) = j, \\ a_j^{(m^*)} & \text{if } j \notin \text{range}(\hat{\sigma}). \end{cases}$$

Decode: $\hat{y} = f_{\text{dec}}(\hat{a})$. Mode 2 is more robust to out-of-distribution inputs since the corpus is used only for imputation, not for the final prediction. At low loss rates Mode 2 is more accurate; at very high loss rates Mode 1 is more robust because imputation quality degrades as more chunks are missing.

5 Theoretical Analysis

We prove that random sign masking creates a reliable positional signal. All results are for the noiseless case first, then extended to noisy channels.

Assumption 3 (Activation model). *After applying the global permutation Π , activation chunks satisfy $\|a_j\|_2 = 1$. For positions $j \neq j'$, the coordinates $\{a_{j,\ell}\}$ and $\{a_{j',\ell}^{(m)}\}$ are approximately coordinate-wise uncorrelated: $\mathbb{E}[a_{j,\ell} a_{j',\ell}^{(m)}] \approx 0$.*

5.1 Cross-Position Analysis: Mean and Variance

Lemma 1 (Zero mean at incorrect positions). *Under Assumption 3, for $j \neq j'$:*

$$\mathbb{E}_{s_j, s_{j'}} [\langle \tilde{a}_j, \tilde{a}_{j'}^{(m)} \rangle] = 0.$$

Proof. Expand the inner product using (2):

$$\langle \tilde{a}_j, \tilde{a}_{j'}^{(m)} \rangle = \sum_{\ell=1}^{d_c} (s_{j,\ell} s_{j',\ell}) a_{j,\ell} a_{j',\ell}^{(m)}.$$

For $j \neq j'$, s_j and $s_{j'}$ are independent, so for each ℓ : $s_{j,\ell} s_{j',\ell} \sim \text{Uniform}\{-1, +1\}$, independent of all activation values. Therefore:

$$\mathbb{E} [\langle \tilde{a}_j, \tilde{a}_{j'}^{(m)} \rangle] = \sum_{\ell} \underbrace{\mathbb{E}[s_{j,\ell} s_{j',\ell}]}_{=0} a_{j,\ell} a_{j',\ell}^{(m)} = 0. \quad \square$$

$\square$

Lemma 2 (Variance at incorrect positions). *Under Assumption 3, for $j \neq j'$:*

$$\text{Var}[\langle \tilde{a}_j, \tilde{a}_{j'}^{(m)} \rangle] = \sum_{\ell=1}^{d_c} a_{j,\ell}^2 (a_{j',\ell}^{(m)})^2 \leq \frac{1}{d_c}.$$

Proof. Since the mean is zero, variance equals the second moment. Letting $Z = \langle \tilde{a}_j, \tilde{a}_{j'}^{(m)} \rangle$ and $\xi_\ell = s_{j,\ell} s_{j',\ell}$:

$$\begin{aligned} \mathbb{E}[Z^2] &= \mathbb{E}\left[\left(\sum_{\ell} \xi_{\ell} a_{j,\ell} a_{j',\ell}^{(m)}\right)^2\right] \\ &= \sum_{\ell} \mathbb{E}[\xi_{\ell}^2] a_{j,\ell}^2 (a_{j',\ell}^{(m)})^2 + \sum_{\ell \neq \ell'} \underbrace{\mathbb{E}[\xi_{\ell}]}_{=0} \underbrace{\mathbb{E}[\xi_{\ell'}]}_{=0} (\dots) \\ &= \sum_{\ell} a_{j,\ell}^2 (a_{j',\ell}^{(m)})^2, \end{aligned} \quad (3)$$

where we used $\mathbb{E}[\xi_{\ell}^2] = 1$ and independence across ℓ . For the upper bound, the global permutation Π spreads activation energy uniformly, so $\max_{\ell} (a_{j',\ell}^{(m)})^2 \leq \|a_{j'}^{(m)}\|_2^2 / d_c = 1/d_c$. Therefore:

$$\sum_{\ell} a_{j,\ell}^2 (a_{j',\ell}^{(m)})^2 \leq \frac{1}{d_c} \sum_{\ell} a_{j,\ell}^2 = \frac{\|a_j\|_2^2}{d_c} = \frac{1}{d_c}. \quad \square$$

□

Remark 1. Theorems 1 and 2 together show that cross-position inner products concentrate around zero with standard deviation $O(1/\sqrt{d_c})$. The correct-match signal $\rho_j = \langle a_j, a_j^{(m^*)} \rangle = O(1)$. The alignment SNR is therefore $\Theta(\rho_j \sqrt{d_c})$, growing as $\sqrt{d_c}$. Larger chunk dimensions make alignment exponentially more reliable.

5.2 Signal Preservation at the Correct Position

Lemma 3 (Correct-position signal). *For correct position j and correct corpus entry m^* , noiseless:*

$$\langle \tilde{a}_j, \tilde{a}_j^{(m^*)} \rangle = \langle a_j, a_j^{(m^*)} \rangle \rho_j > 0.$$

Proof. $\langle \tilde{a}_j, \tilde{a}_j^{(m^*)} \rangle = \sum_{\ell} s_{j,\ell}^2 a_{j,\ell} a_{j,\ell}^{(m^*)} = \sum_{\ell} a_{j,\ell} a_{j,\ell}^{(m^*)} = \langle a_j, a_j^{(m^*)} \rangle$, since $s_{j,\ell}^2 = 1$. □ □

5.3 Recovery Guarantee Under Noise

Theorem 4 (Recovery condition). *Under ?? 1–3 with $\Sigma_n = \sigma^2 I$, the probability that Stage 2 of Algorithm 1 assigns all $P - k$ received packets to their correct positions satisfies:*

$$\mathbb{P}[\text{correct alignment}] \geq 1 - P^2 \exp\left(-\frac{c d_c \Delta^2}{1 + d_c \sigma^2}\right),$$

where $\Delta = \min_j \rho_j$ is the minimum correct-match signal across chunk positions and $c > 0$ is an absolute constant.

Proof. Fix received packet q_i with true position j and true corpus entry m^* . Correct assignment requires:

$$B_{ij} = \langle q_i, \tilde{a}_j^{(m^*)} \rangle > B_{ij'} \quad \forall j' \neq j.$$

Step 1: Score decomposition. Write $q_i = \tilde{a}_j + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2 I)$.

$$B_{ij} = \rho_j + \eta_j, \quad \eta_j = \langle \epsilon_i, \tilde{a}_j^{(m^*)} \rangle \sim \mathcal{N}(0, \sigma^2). \quad (4)$$

$$B_{ij'} = Z_{jj'} + \eta_{j'}, \quad \mathbb{E}[Z_{jj'}] = 0, \quad \text{Var}[Z_{jj'}] \leq 1/d_c, \quad \eta_{j'} \sim \mathcal{N}(0, \sigma^2). \quad (5)$$

Step 2: Concentration of $Z_{jj'}$. Each term in $Z_{jj'} = \sum_{\ell} \xi_{\ell} a_{j,\ell} a_{j',\ell}^{(m^*)}$ is bounded in $[-1/d_c, 1/d_c]$ after normalization. By Hoeffding’s inequality:

$$\mathbb{P}[Z_{jj'} \geq t] \leq \exp(-2d_c t^2).$$

The total score $B_{ij'}$ has sub-Gaussian parameter $\tau^2 = 1/(2d_c) + \sigma^2/2$, giving:

$$\mathbb{P}[B_{ij'} \geq t] \leq \exp\left(-\frac{d_c t^2}{2(1 + d_c \sigma^2)}\right).$$

Step 3: Union bound over incorrect positions. Setting $t = \Delta/2$ and applying a union bound over $P - 1$ incorrect positions:

$$\mathbb{P}\left[\max_{j' \neq j} B_{ij'} \geq \frac{\Delta}{2}\right] \leq (P - 1) \exp\left(-\frac{d_c \Delta^2}{8(1 + d_c \sigma^2)}\right).$$

The probability that $\eta_j \leq -\Delta/2$ (correct score falls below threshold) is $\leq \exp(-\Delta^2/(8\sigma^2))$. Combining:

$$\mathbb{P}[\text{error at packet } i] \leq P \exp\left(-\frac{c d_c \Delta^2}{1 + d_c \sigma^2}\right).$$

Step 4: Union bound over all packets.

$$\mathbb{P}[\text{any error}] \leq (P - k) \cdot P \exp\left(-\frac{c d_c \Delta^2}{1 + d_c \sigma^2}\right) \leq P^2 \exp\left(-\frac{c d_c \Delta^2}{1 + d_c \sigma^2}\right). \quad \square$$

$\square$

Corollary 5 (Minimum chunk dimension for reliable recovery). *For target error probability δ :*

$$d_c \geq \frac{2 \log(P^2/\delta)}{c \Delta^2 - \sigma^2 \log(P^2/\delta)}$$

guarantees $\mathbb{P}[\text{correct alignment}] \geq 1 - \delta$. Recovery becomes easier as Δ increases and harder as σ^2 or P increases.

Corollary 6 (High loss rate and Stage 1). *At high loss rates, fewer packets contribute to $\text{score}(m)$ in Stage 1. The per-packet alignment guarantee of Theorem 4 is unchanged, but the probability that Stage 1 correctly identifies m^* decreases. At high p_{loss} , corpus identification is the binding constraint, not position assignment.*

6 Optional: Alignment-Aware Fine-Tuning

The method operates zero-shot on any pretrained model. For very low channel SNR, an optional contrastive fine-tuning stage can shift the recovery threshold of Theorem 4 by increasing Δ . We fine-tune f_{enc} with InfoNCE [van den Oord et al., 2018] at the chunk level:

$$\mathcal{L}_{\text{align}} = -\frac{1}{P} \sum_{j=1}^P \log \frac{\exp(\langle \tilde{a}_j, \tilde{a}_j^+ \rangle / T)}{\sum_{j'=1}^P \exp(\langle \tilde{a}_j, \tilde{a}_{j'}^+ \rangle / T)},$$

where $\tilde{a}_j^+$ is the masked corpus chunk at position j for the same input. Positive pairs: (chunk, same-position corpus chunk). Negatives: all other positions. Only f_{enc} is modified, not f_{dec} . This is substantially lighter than SI-NR [Shao et al., 2023], which retrains end-to-end with a separate model per loss regime.

7 Experiments

7.1 Setup

Models. Vision: MobileNetV2, EfficientNet-B0, EfficientNet-B4 (torchvision), ResNet50 (torchvision), DeiT-Small (timm), ViT-B16 (torchvision). Language: DistilBERT-base-uncased and BERT-Base-uncased (HuggingFace). All models use pretrained weights with no fine-tuning.

Table 1: MobileNetV2 on ImageNet-1K, permuted (loss + reordering).

Condition	p_{loss}	Top-1	Top-5	CE
Clean	0%	63.0	85.0	1.678
Oracle	20%	58.0	81.0	1.900
Rematched	20%	58.0	81.0	1.900
Random	20%	8.0	20.2	5.672
Oracle	40%	45.0	77.0	2.288
Rematched	40%	45.0	77.0	2.288
Random	40%	2.0	5.4	6.813

Table 2: ResNet50 on ImageNet-1K, permuted (loss + reordering).

Condition	p_{loss}	Top-1	Top-5	CE
Clean	0%	76.0	97.0	1.419
Oracle	20%	72.0	95.0	2.179
Rematched	20%	72.0	95.0	2.179
Random	20%	21.2	40.4	6.436
Oracle	40%	71.0	94.0	3.253
Rematched	40%	71.0	94.0	3.253
Random	40%	8.2	15.8	6.719

Datasets. ImageNet-1K (50,000 validation images, 1,000 classes). GLUE: SST-2 (872 validation sentences) and MNLI (9,815 validation pairs).

Transmission modes. *Contiguous*: packets sent in order (loss only). *Permuted*: chunks shuffled uniformly at random before packing (loss + full reordering). The permutation is unknown to the receiver.

Packet loss rates. $p_{\text{loss}} \in \{0.20, 0.40, 0.60, 0.70, 0.80\}$.

Corpus. $M = 1000$ activations from the training set unless noted.

7.2 Baselines

Clean: no loss, no reordering. **Oracle**: true positions known; lost chunks zeroed — cost of loss alone. **Random**: positions assigned uniformly at random — cost of misalignment. **Rematched (ours)**: Algorithm 1 with Hungarian matching.

7.3 Results

Tables 1–3 report results for the permuted setting (simultaneous loss and reordering).

7.4 Key Observations

Rematched equals oracle exactly. Across all tested architectures and loss rates, under full reordering, rematched accuracy is numerically identical to oracle. Residual degradation from clean is due entirely to zeroed lost packets, not misalignment.

Gap over random is extreme. At 40% loss on ViT-B16: random achieves 0% Top-1; rematched achieves 79%. Vision transformers rely more heavily on positional structure than CNNs and benefit most from correct alignment.

Alignment introduces zero error. The 1 percentage-point gap between rematched (79%) and clean (80%) at 40% loss on ViT-B16 is entirely attributable to missing packets, not misalignment.

Table 3: ViT-B16 on ImageNet-1K, permuted (loss + reordering).

Condition	p_{loss}	Top-1	Top-5	CE
Clean	0%	80.0	96.0	0.752
Oracle	20%	81.0	97.0	0.784
Rematched	20%	81.0	97.0	0.784
Random	20%	0.0	0.2	7.271
Oracle	40%	79.0	94.0	0.970
Rematched	40%	79.0	94.0	0.970
Random	40%	0.0	0.4	7.355

8 Ablation Studies

Ablation 1 — Phase transition [Theorem 4]. All models at $p_{\text{loss}} \in \{0.60, 0.70, 0.80\}$. Plot Top-1 vs. p_{loss} for {clean, oracle, rematched, random}. The point where rematched diverges from oracle is the empirical phase transition; compare against the theoretical threshold from Theorem 5.

Ablation 2 — Chunk dimension [Theorem 5]. Fix d ; vary $P \in \{2, 4, 8, 16, 32\}$ giving $d_c = d/P$. Plot chunk-level alignment accuracy vs. d_c . Verify exponential concentration.

Ablation 3 — Corpus size [Theorem 6]. Vary $M \in \{100, 500, 1000, 5000, 10000\}$. Plot retrieval accuracy ($\mathbb{P}[m^* = m_{\text{true}}]$) and Top-1 vs. M . Report saturation point.

Ablation 4 — Greedy vs. Hungarian [Section 4.4.2]. Compare at $p_{\text{loss}} \in \{0.4, 0.6, 0.7, 0.8\}$. Report accuracy gap and latency ratio. Identify the loss rate at which greedy diverges.

Ablation 5 — Split layer depth. ResNet50 and ViT-B16: vary split from early to late layers. Plot alignment accuracy vs. layer index. Later layers should have higher Δ .

Ablation 6 — Language model generalization. DistilBERT (SST-2) and BERT-Base (MNLI) at $p_{\text{loss}} \in \{0.20, 0.40, 0.60\}$. Report task accuracy and alignment accuracy.

Ablation 7 — Temperature sensitivity. Soft-assignment variant: sweep $T \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$. Report alignment accuracy vs. T .

9 Discussion

The corpus assumption. A server-side corpus is standard in edge inference systems where the server has access to training or validation data. The corpus is computed once offline and does not grow with inference requests. Ablation 3 characterizes the minimum M required.

Out-of-distribution inputs. Mode 1 (retrieval-only) will fail when x is far from all corpus entries. Mode 2 (hybrid) is more robust as it uses the corpus only for imputation. We recommend Mode 2 for open-world deployment.

Sign masking vs. metric-based approaches. An alternative strategy for improving alignment accuracy is to optimize the similarity metric used for matching. Under anisotropic channel noise $\Sigma_n \neq \sigma^2 I$, replacing the standard dot product with the Mahalanobis similarity $q^\top \Sigma_n^{-1} c$ maximizes the alignment SNR (by a Cauchy-Schwarz argument on the matrix inner product, $A^* \propto \Sigma_n^{-1}$). However, noise whitening requires estimating Σ_n from held-out data, adds a matrix-vector multiply at inference time, and provides no benefit when the noise is isotropic — the common case for most channel models. More critically, whitening optimizes the *geometry* of the similarity function but cannot create the mean-zero cross-position structure that is the fundamental property enabling position recovery. Sign masking provides this structural property directly, with no parameter estimation and only an elementwise multiply overhead. The two approaches are complementary: noise whitening can be layered on top of sign masking when Σ_n is known and anisotropic, but sign masking is the necessary and sufficient ingredient for metadata-free alignment.

Representation vs. communication structure. Our method demonstrates that positional metadata is not a necessary ingredient of reliable inference when activations are sufficiently high-dimensional.

This principle may extend to distributed training over unreliable networks and federated learning settings.

10 Conclusion

We have presented a framework for metadata-free split inference under simultaneous packet loss and reordering. Random sign masking embeds chunk position into the representation without any transmitted side information. Corpus identification via aggregation scoring followed by bipartite matching recovers the injective position assignment in $O(MP^2 + P^3)$ time, reducing a search space of size $P!/(P - k)!$ to polynomial time. Theoretical analysis shows that cross-position inner products concentrate around zero with variance $1/d_c$, yielding an explicit recovery guarantee that tightens exponentially with chunk dimension. Empirically, rematched accuracy equals oracle across all tested architectures at up to 40% packet loss under full reordering, with near-zero alignment-induced degradation. Representation structure can replace communication structure as the source of alignment in distributed inference systems.

References

- John J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *PNAS*, 79(8):2554–2558, 1982.
- Yiping Kang, Johann Hauswald, Cao Gao, Austin Rovinski, Trevor Mudge, Jason Mars, and Lingjia Tang. Neurosurgeon: Collaborative intelligence between the cloud and mobile edge. In *ASPLOS*, 2017.
- Harold W Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2:83–97, 1955.
- Patrick Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020.
- En Li, Liekang Zeng, Zhi Zhou, and Xu Chen. JALAD: Joint accuracy- and latency-aware deep structure decoupling for edge-cloud inference. In *ICPADS*, 2018.
- Hubert Ramsauer et al. Hopfield networks is all you need. *arXiv:2008.02217*, 2020.
- Jiawei Shao and Jun Zhang. BottleNet++: An end-to-end approach for feature compression in device-edge co-inference systems. In *IEEE ICC*, 2020.
- Jiawei Shao et al. Wireless deep video semantic transmission. *IEEE Journal on Selected Areas in Communications*, 2023.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv:1807.03748*, 2018.
- Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.

A Complete Proof of Theorem 4

We reproduce all steps with explicit constants.

Setting. Packet q_i with true position j , true corpus m^* , noise $\epsilon_i \sim \mathcal{N}(0, \sigma^2 I)$.

Step 1: Decompose scores. $B_{ij} = \rho_j + \eta_j$, $\eta_j \sim \mathcal{N}(0, \sigma^2)$. $B_{ij'} = Z_{jj'} + \eta_{j'}$, $\mathbb{E}[Z_{jj'}] = 0$, $\text{Var}[Z_{jj'}] \leq 1/d_c$, $\eta_{j'} \sim \mathcal{N}(0, \sigma^2)$.

Step 2: Hoeffding on $Z_{jj'}$. Each summand of $Z_{jj'}$ is bounded in $[-|a_{j,\ell} a_{j',\ell}^{(m^*)}|, +|a_{j,\ell} a_{j',\ell}^{(m^*)}|]$. By Hoeffding: $\mathbb{P}[Z_{jj'} \geq t] \leq \exp(-2d_c t^2)$.

Step 3: Combined tail bound. The total score $B_{ij'} = Z_{jj'} + \eta_{j'}$ has combined variance $\tau^2 \leq 1/(2d_c) + \sigma^2/2$. By sub-Gaussian tail bounds:

$$\mathbb{P}[B_{ij'} \geq t] \leq \exp\left(-\frac{d_c t^2}{2(1 + d_c \sigma^2)}\right).$$

Step 4: Set threshold. With $t = \Delta/2$:

$$\mathbb{P}\left[\max_{j' \neq j} B_{ij'} \geq \frac{\Delta}{2}\right] \leq (P-1) \exp\left(-\frac{d_c \Delta^2}{8(1 + d_c \sigma^2)}\right).$$

Step 5: Signal falls below threshold. $\mathbb{P}[\eta_j \leq -\Delta/2] \leq \exp(-\Delta^2/(8\sigma^2))$.

Step 6: Per-packet error. Combining Steps 4 and 5 with a union bound: $\mathbb{P}[\text{error at } i] \leq P \exp(-cd_c \Delta^2/(1 + d_c \sigma^2))$ where $c = 1/8$.

Step 7: Union over all packets. $\mathbb{P}[\text{any error}] \leq P^2 \exp(-cd_c \Delta^2/(1 + d_c \sigma^2))$. $\square$

B Complexity Analysis

Stage	Method	Complexity
Corpus scoring	Aggregation	$O(MP^2)$
Assignment	Hungarian	$O(P^3)$
Assignment	Greedy	$O(P^2)$
Unmasking	Elementwise multiply	$O(d)$
Imputation	Copy from corpus	$O(k d_c)$
Total (Hungarian)		$O(MP^2 + P^3)$
Total (Greedy)		$O(MP^2 + P^2)$

For $M = 1000$, $P = 16$, $d_c = 64$: aggregation requires $\sim 10^7$ multiply-accumulates; Hungarian on a 15×16 matrix takes $\ll 0.1$ ms on CPU.

C Ablation Implementation Guide

This section provides pseudocode for the code agent implementing each ablation.

C.1 Phase Transition (Ablation 1)

```

1: for model in [MobileNetV2, EfficientNet-B0, B4, ResNet50, DeiT-Small, ViT-B16, DistilBERT, BERT-Base] do
2:   Build masked corpus  $\mathcal{C}$  from training set,  $M = 1000$ 
3:   for  $p_{\text{loss}}$  in [0.0, 0.2, 0.4, 0.6, 0.7, 0.8] do
4:     for each validation example  $x$  do
5:       Compute, permute, chunk, mask  $\rightarrow \{\tilde{a}_j\}$ 
6:       Drop each  $\tilde{a}_j$  w.p.  $p_{\text{loss}}$ ; shuffle survivors
7:       Run Algorithm 1; reconstruct via Mode 2; decode
8:       Record Top-1, Top-5, CE loss, retrieval accuracy
9:     end for
10:   end for
11:   Plot Top-1 vs.  $p_{\text{loss}}$ ; mark empirical phase transition; overlay Theorem 5 threshold
12: end for
```

C.2 Chunk Dimension Sweep (Ablation 2)

```

1: for model in [ResNet50, ViT-B16] do
2:   for  $P$  in [2, 4, 8, 16, 32],  $d_c = d/P$  do
3:     Regenerate  $\{s_j\}_{j=1}^P \in \{-1, +1\}^{d_c}$ 
4:     for  $p_{\text{loss}}$  in [0.4, 0.6] do
```

```

5:         Run alignment on 1000 validation examples
6:         Record chunk-level alignment accuracy
7:     end for
8: end for
9: Plot accuracy vs.  $d_c(\log-x)$ ; overlay  $1 - P^2 \exp(-cd_c\Delta^2)$  with estimated  $\Delta$ 
10: end for

```

C.3 Corpus Size Sweep (Ablation 3)

```

1: for  $M$  in [100, 500, 1000, 5000, 10000, 50000] do
2:   for  $p_{\text{loss}}$  in [0.2, 0.4, 0.6] do
3:     Run alignment on 1000 examples
4:     Record retrieval accuracy  $\mathbb{P}[m^* = m_{\text{true}}]$  and Top-1
5:   end for
6: end for
7: Plot both metrics vs.  $M(\log-x)$ ; report saturation point

```

C.4 Greedy vs. Hungarian (Ablation 4)

```

1: for model in [ResNet50, ViT-B16] do
2:   for  $p_{\text{loss}}$  in [0.4, 0.6, 0.7, 0.8] do
3:     Run Hungarian variant; record (accuracy, wall-clock latency)
4:     Run Greedy variant; record (accuracy, wall-clock latency)
5:   end for
6: end for
7: Report accuracy gap (Hungarian – greedy) vs.  $p_{\text{loss}}$ 
8: Report latency ratio (Hungarian / greedy) vs.  $p_{\text{loss}}$ 

```

Place the following in `references.bib`:

```

@inproceedings{kang2017neurosurgeon,
  title={Neurosurgeon: Collaborative Intelligence Between
    the Cloud and Mobile Edge},
  author={Kang, Yiping and Hauswald, Johann and Gao, Cao
    and Rovinski, Austin and Mudge, Trevor and
    Mars, Jason and Tang, Lingjia},
  booktitle={ASPLoS},
  year={2017}
}

@inproceedings{shao2020bottlenet,
  title={{BottleNet++}: An End-to-End Approach for Feature
    Compression in Device-Edge Co-Inference Systems},
  author={Shao, Jiawei and Zhang, Jun},
  booktitle={IEEE ICC},
  year={2020}
}

@inproceedings{li2018jalad,
  title={{JALAD}: Joint Accuracy- and Latency-Aware Deep
    Structure Decoupling for Edge-Cloud Inference},
  author={Li, En and Zeng, Liekang and Zhou, Zhi
    and Chen, Xu},
  booktitle={ICPADS},
  year={2018}
}

@article{shao2023wireless,
  title={Wireless Deep Video Semantic Transmission},
  author={Shao, Jiawei and others},
  journal={IEEE Journal on Selected Areas in

```

```

        Communications},
    year={2023}
}

@inproceedings{lewis2020rag,
    title={Retrieval-Augmented Generation for
        Knowledge-Intensive {NLP} Tasks},
    author={Lewis, Patrick and others},
    booktitle={NeurIPS},
    year={2020}
}

@article{oord2018cpc,
    title={Representation Learning with Contrastive
        Predictive Coding},
    author={van den Oord, Aaron and Li, Yazhe
        and Vinyals, Oriol},
    journal={arXiv:1807.03748},
    year={2018}
}

@article{hopfield1982,
    title={Neural Networks and Physical Systems with
        Emergent Collective Computational Abilities},
    author={Hopfield, John J},
    journal={PNAS},
    volume={79},
    number={8},
    pages={2554--2558},
    year={1982}
}

@article{ramsauer2020hopfield,
    title={Hopfield Networks Is All You Need},
    author={Ramsauer, Hubert and others},
    journal={arXiv:2008.02217},
    year={2020}
}

@book{vershynin2018hdp,
    title={High-Dimensional Probability: An Introduction
        with Applications in Data Science},
    author={Vershynin, Roman},
    publisher={Cambridge University Press},
    year={2018}
}

@article{kuhn1955hungarian,
    title={The {Hungarian} Method for the Assignment
        Problem},
    author={Kuhn, Harold W},
    journal={Naval Research Logistics Quarterly},
    volume={2},
    pages={83--97},
    year={1955}
}

```